

JORDI FABREGAT

Gameplay Programmer · Unreal Engine 5 / C++ · Unity / C# · Technical Designer

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PROFILE

Computer Engineering graduate (UPF, June 2026) who ships playable systems. Currently building **Do Not Follow**, a first-person horror game in Unreal Engine 5 / C++, and recently launched **Axiom Arena**, a live competitive math game with real-time 1v1 multiplayer over an authoritative server. Four shipped projects across UE5, Unity, a C++ engine framework and the web, plus six months in a research lab iterating mixed-reality games with scientists. I own a feature from design document to balanced, shippable build. Available immediately.

SELECTED PROJECTS

Do Not Follow — Solo · Unreal Engine 5 / C++ 2026 – present

First-person investigation horror game. Vertical slice with the full mechanic set complete; in development.

- Implemented every gameplay system in Unreal C++: tiered environmental-storytelling lore, a symbol-warding puzzle, a fuse-for-darkness risk/reward trade-off, AI-summoning "lullaby" zones and a possession-climax door duel.
- Built and tuned the enemy threat on sight + hearing AI perception so the safe/danger contrast that dread depends on holds across explorable hubs and channelled connectors.
- Authored the design frame (dread over chase), spatial model and lore bible that drive the level's progression.

Axiom Arena — Solo · TypeScript, React, Node.js, WebSockets, PostgreSQL (Supabase) 2026 · live at axiombattlearena.com

Competitive math game: 60-second 1v1 duels, ranked ladder, story mode, collectible cards. Launched September 2026.

- Designed and shipped the whole product: rating-based matchmaking, Elo ladder with tiers, divisions and seasons, quest-line story mode, card packs with published pull odds, coins, shop and accounts.
- Architected an authoritative game server: question generation, clocks, scoring and rewards run server-side so clients cannot forge results; one pure rules module is shared by client and server.
- CI with headless-browser end-to-end tests (forged actions rejected, full duels scored server-side); deployed on Railway + Netlify with Supabase auth.

4-player co-op heist game — Solo · pre-production 2026

- Designing an online co-op heist where four animal thieves loot guarded warehouses, compete for the biggest haul while needing a shared minimum cut to progress, then spend it on upgrades and ways to sabotage each other.

Alphinland — Solo · Unity / C# 2026 · itch.io

- First-person narrative walking sim about grief, built end to end in one month: three-tier interaction system (info / pickup / inspect), a custom dual-world shader transition tied to story beats, and a diegetic "grief meter" that gates progression with no HUD explanation.

Dual Forts — Team · Unity / C# · HTC Vive trackers 2025

- Co-op mixed-reality tower defense for a 6×6 m projected playspace. Implemented the wave, tower and enemy systems and led the balancing pass (HP curves, tower pricing, wave pacing); designed the forced-cooperation resource mechanic.

Device Bound — Solo · C++ on a partial engine framework 2025

- First-person puzzle-platformer on a bare engine skeleton: completed the renderer, added frustum culling and Phong shading with distance fog; throw-to-teleport mechanic with three throw velocities, bounce banking and save/rewind. Published by faculty as a reference example for later cohorts (9/10).

EXPERIENCE

Gameplay Programmer / Technical Designer (R&D Assistant) — Full-Body Interaction Lab, UPF Sep 2025 – Feb 2026

- Iterated the Unity / C# codebase of *DragonIce* and *Lands of Fog*, full-body mixed-reality games used in published HCI research with neurodivergent children (Lands of Fog is funded by RecerCaixa).
- Fixed the stability and logging issues that had been blocking clean behavioural data capture; shipped weekly iteration cycles with PhD researchers and supported on-site sessions.

EDUCATION

BSc Computer Engineering — Universitat Pompeu Fabra, Barcelona 2022 – Jun 2026

- Specialisation: Video Game Development & Interaction. Thesis *Technical and Strategic Analysis of Video Game Success* (10/10): why AAA and indie games succeed or fail, from analysis of thousands of Steam reviews.
- Coursework: Game Engine Development (C++), Interactive Systems, Human-Computer Interaction, Narrative & Videogames, Data Structures & Algorithms, Databases.

SKILLS

Languages: C++ · C# · TypeScript / JavaScript · Python · SQL · Java

Engines & tools: Unreal Engine 5 (C++, AI perception) · Unity · React · Node.js · WebSockets · PostgreSQL / Supabase · Git · GitHub Actions CI · Visual Studio

Gameplay: systems & interaction design · AI behaviour and pacing · multiplayer architecture (authoritative server, matchmaking, Elo) · balancing · shaders · rapid prototyping · playtesting · GDD authoring

Spoken: English C2 · Spanish native · Catalan native